

Seasonal and Geospatial Analysis on Geochemical Ocean Conditions in Areas with High Marine Phytoplankton Productivity

Anna Chen | Lamont-Doherty Earth Observatory, Columbia University | PI: Dr. Sarah Hurley; Dr. Henry Holm | Summer 2024

Background

Marine phytoplankton fix carbon by converting CO₂ into O₂, which can be impacted by global warming and ocean acidification.

Global phytoplankton growth (Net Primary Productivity) can be estimated using satellite data through a Carbon-based Production Model (CbPM).

Among all pelagic physical conditions, temperature and partial pressure of CO₂ (pCO₂) levels play major roles in determining phytoplankton NPP.

Guiding Question: What are the physical ocean conditions in areas with high phytoplankton growth, and how do they differ from average conditions?

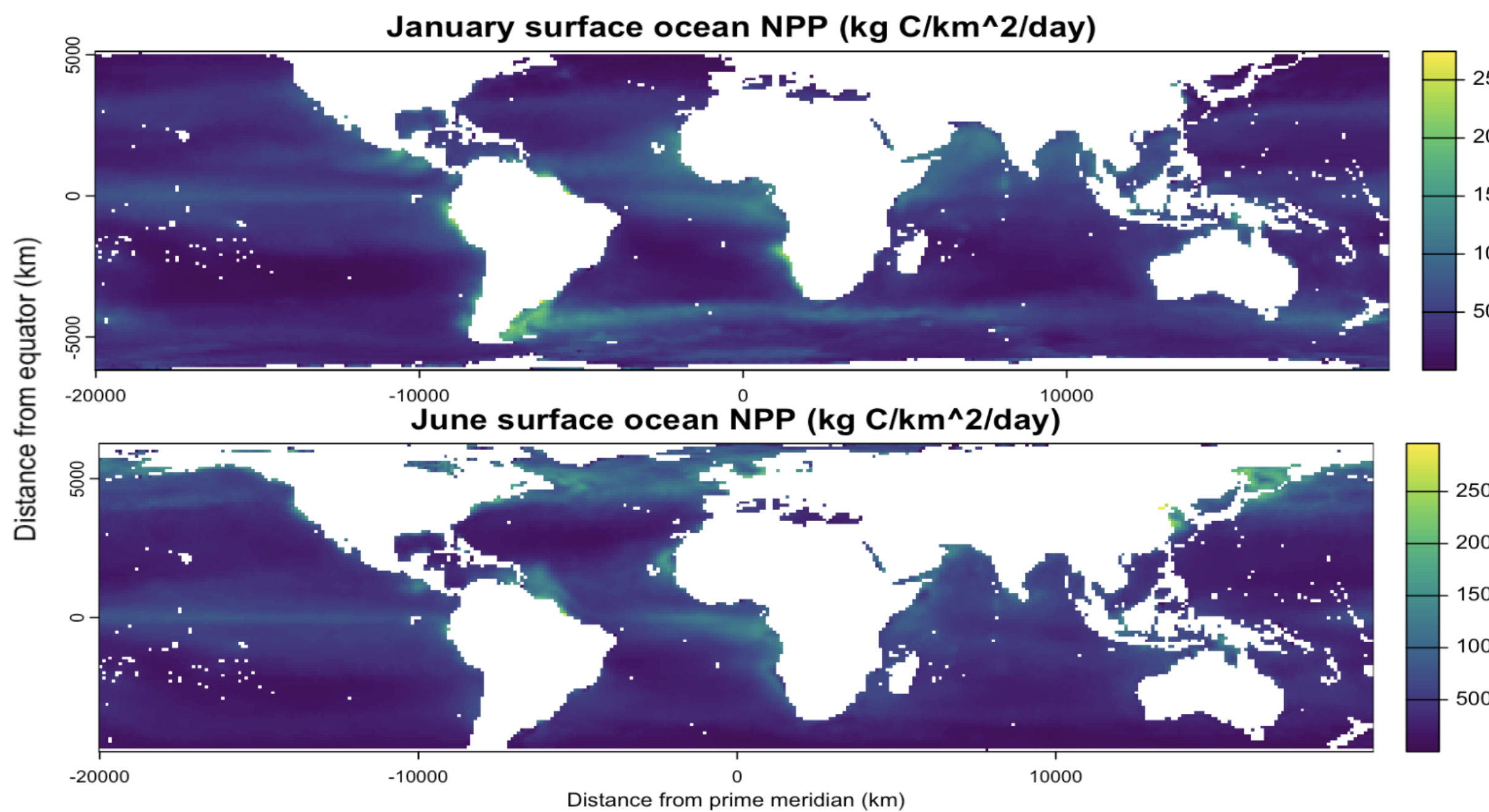
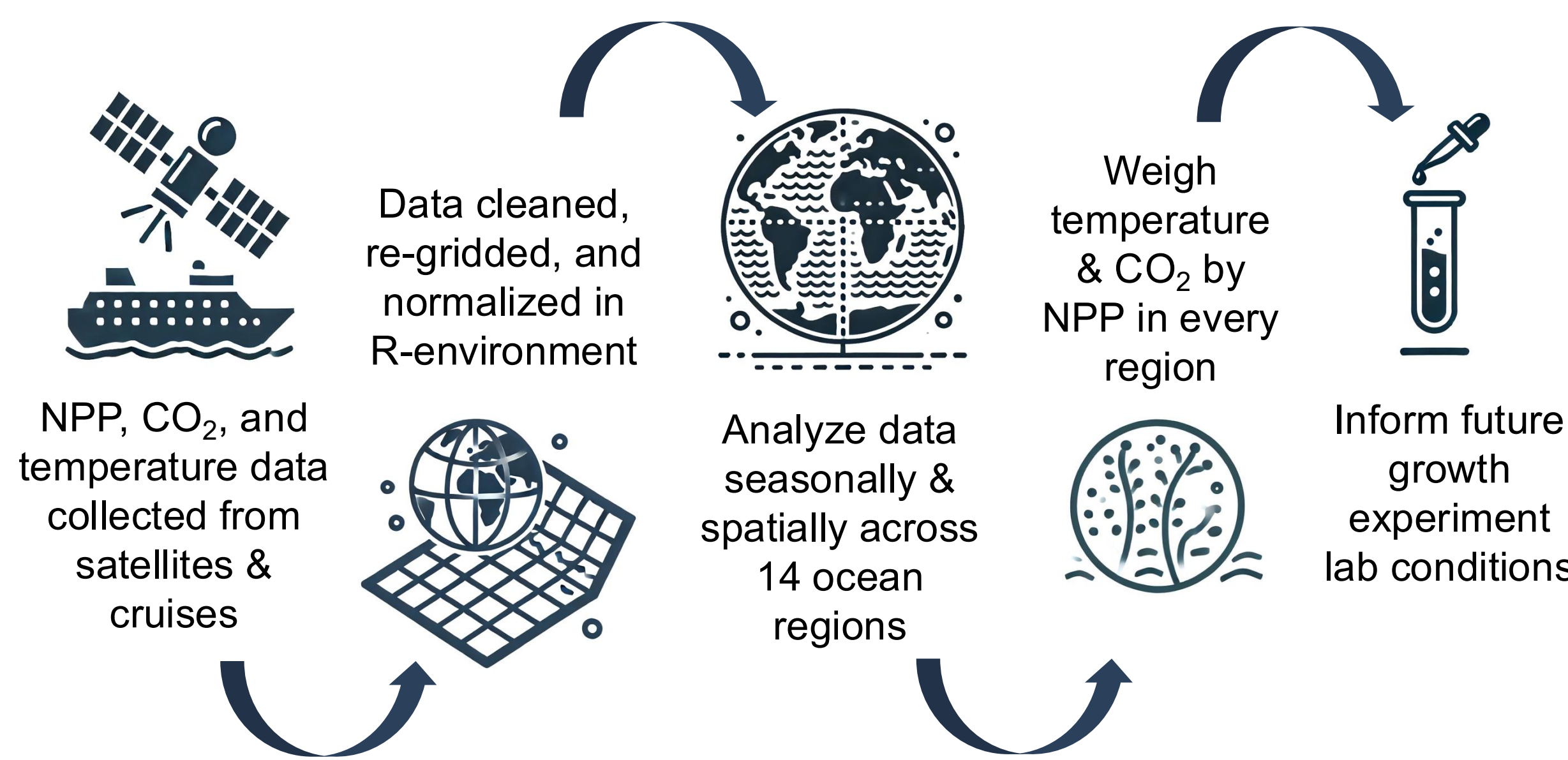


Fig. 1: Seasonally-varied global surface ocean Net Primary Productivity, re-gridded and re-projected

Methodology



Results

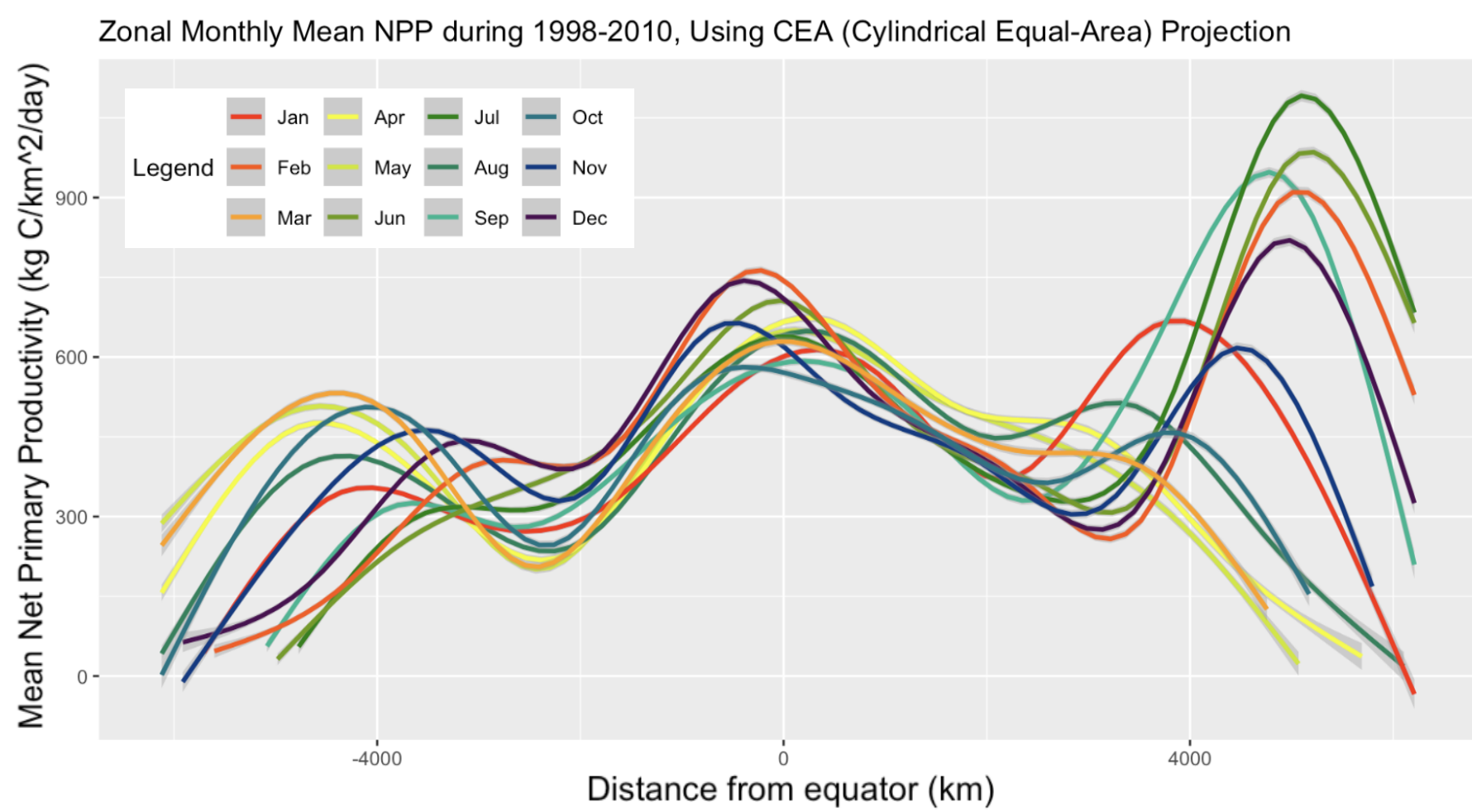


Fig. 2: Monthly-varied zonal mean Net Primary Production

Equator NPP peak is seasonally constant

High-latitude (~45°) NPP peak is more seasonally dependent, with summer blooms being much more significant

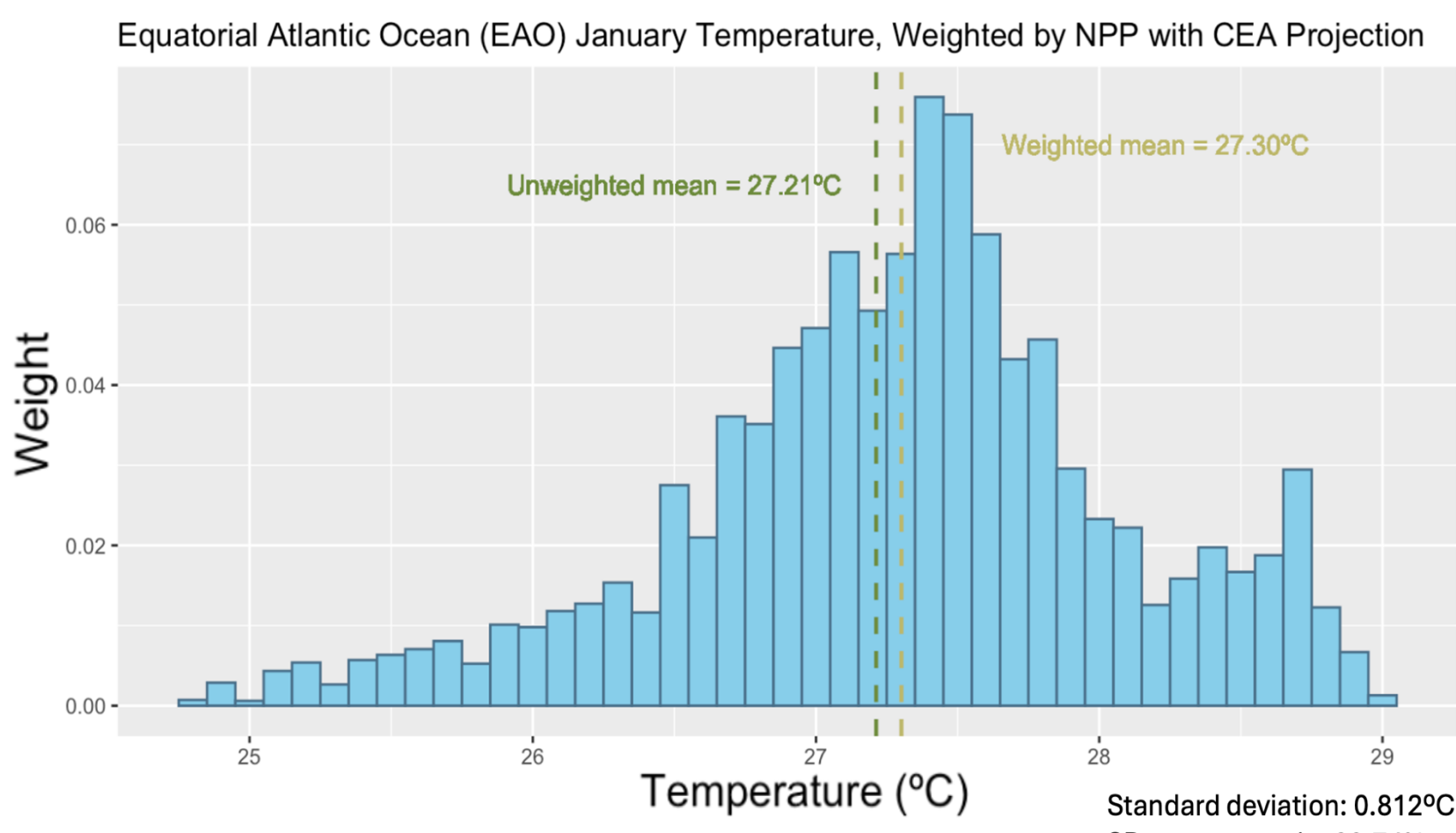


Fig. 3a: Temperature weighted by NPP in Equatorial Atlantic

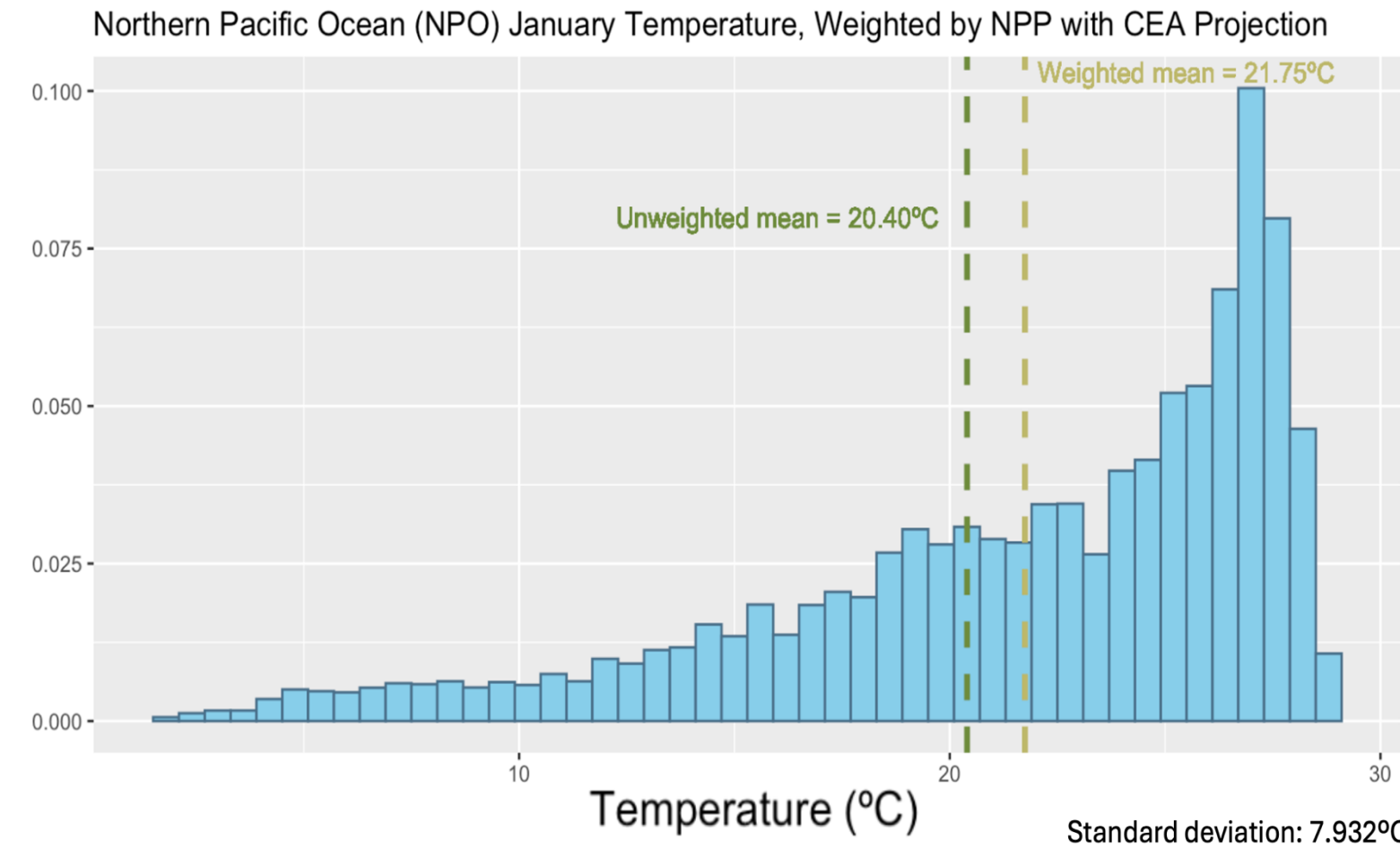
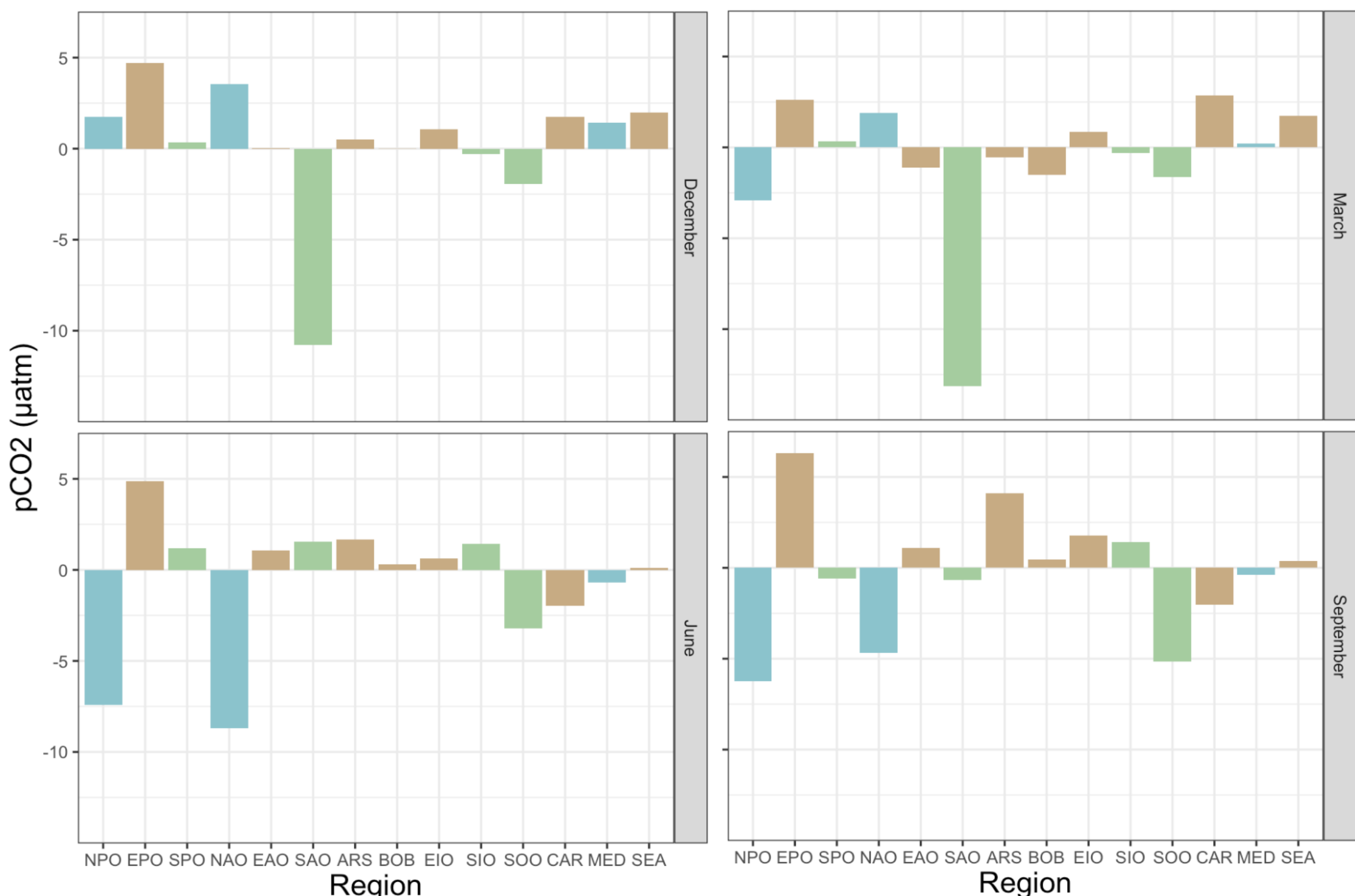
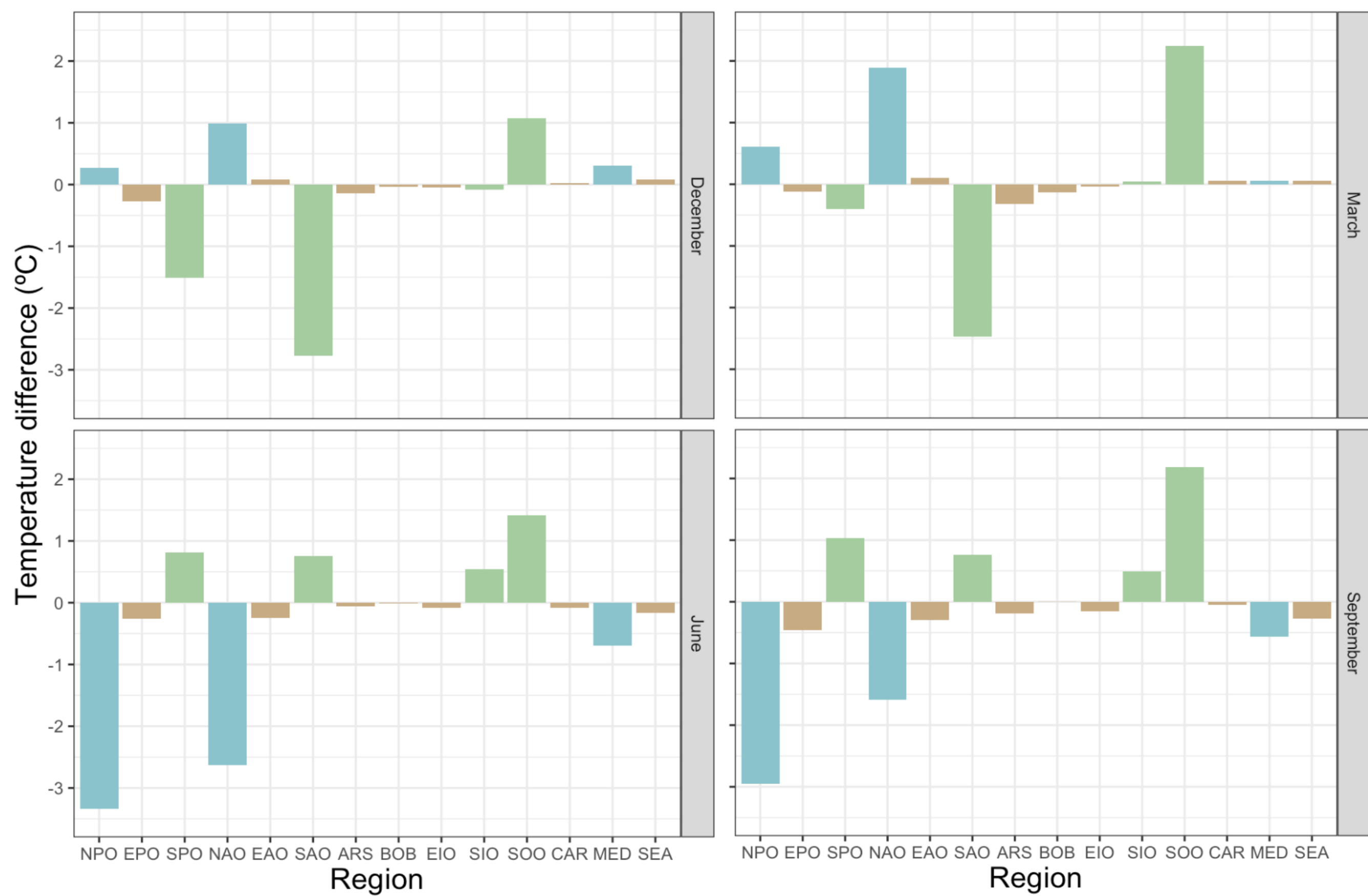


Fig. 3b: Temperature weighted by NPP in Northern Pacific

Temperature and pCO₂ weighted by NPP to show the conditions most relevant to phytoplankton growth of the given region and season

In situ (unweighted) temperature generally lower than both growth-weighted temperature and pCO₂

Larger differences between weighted and unweighted means and larger uncertainties in high-latitude regions



(Left) Fig. 4a: Regional deviation between temperature weighted by NPP and in-situ temperature

(Right) Fig. 4b: Regional deviation between pCO₂ weighted by NPP and in-situ pCO₂

Conclusion

The physical conditions of areas with high marine net primary production (NPP) vary seasonally and latitudinally, especially in high latitudes.

Summer blooms occur in temperatures and pCO₂ conditions lower than regional averages, and winter vice versa.

Peak latitude range, magnitude, and physical conditions of equatorial phytoplankton growth are relatively constant and have lower uncertainties.

Further implications: for lab-based experiments comparable to growing plankton, focus on deviations between high-growth area conditions and averages can reflect temporally and spatially varied growth, helping to set lab experiment parameters.

Acknowledgments

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